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Imaging assistance apparatus, imaging system, imaging assistance method, and imaging assistance program

Abstract

In one aspect, an imaging assistance apparatus includes: a laser light projection device having a laser light source that generates laser light and a laser head provided with an optical element that projects the laser light as patterned light having a determined pattern; a reception device that receives a first operation; a projection instruction device that instructs the laser light projection device during laser light projection to end the projection of the patterned light, on the basis of the first operation; and an imaging instruction device that instructs an imaging device to image a subject on which focusing control is performed, on the basis of the first operation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of PCT International Application No. PCT/JP2021/001039 filed on Jan. 14, 2021, which claims priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2020-064082 filed on Mar. 31, 2020. Each of the above application(s) is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates to a technique of imaging a subject by performing focusing

control of an imaging device by using auxiliary light.

2. Description of the Related Art

(2) For example, JP2002-236251A discloses that laser light is used as AF auxiliary light. Further, JP2009-086559A discloses that a focus position is set by using a laser pointer.

SUMMARY OF THE INVENTION

(3) However, the technique disclosed in JP2002-236251A described above requires a hologram plate. Further, in the technique disclosed in JP2009-086559A, the laser pointer irradiates a single light spot.

(4) The present invention has been made in view of such circumstances, and provides an imaging assistance apparatus, an imaging system, an imaging assistance method, and an imaging assistance program capable of satisfactorily performing focusing control by using patterned light.

(5) According to a first aspect of the present invention, there is provided an imaging assistance apparatus comprising: a laser light projection device having a laser light source that generates laser light and a laser head provided with an optical element that projects the laser light as patterned light having a determined pattern; a reception device that receives a first operation; a projection instruction device that instructs the laser light projection device during laser light projection to end the projection of the patterned light, on the basis of the first operation; and an imaging instruction device that instructs an imaging device to image a subject on which focusing control is performed, on the basis of the first operation.

(6) In the imaging assistance apparatus according to a second aspect, in the first aspect, the reception device may receive a preparation operation before the first operation, the projection instruction device may instruct the laser light projection device to start the projection of the patterned light, on the basis of the preparation operation, and a focusing instruction device that instructs the imaging device to start the focusing control on the subject on which the patterned light is projected, on the basis of the preparation operation, may be further provided.

(7) In the imaging assistance apparatus according to a third aspect, in the first aspect, the projection instruction device may instruct the laser light projection device to start the projection of the patterned light, on the basis of the first operation, and a focusing instruction device that instructs the imaging device to start the focusing control on the subject on which the patterned light is projected, on the basis of the first operation, may be further provided.

(8) In the imaging assistance apparatus according to a fourth aspect, in the third aspect, the projection instruction device may instruct the laser light projection device to end the projection of the patterned light at a fixed interval after the start of the projection of the patterned light, on the basis of the first operation, and the imaging instruction device may instruct the imaging device to image the subject on which the focusing control is performed, after the focusing instruction device instructs the imaging device to start the focusing control, on the basis of the first operation.

(9) In the imaging assistance apparatus according to a fifth aspect, in any one of the first to fourth aspects, the reception device may receive a second operation, and the projection instruction device may instruct the laser light projection device to change a projection direction of the patterned light, according to the second operation.

(10) In the imaging assistance apparatus according to a sixth aspect, in any one of the first to fifth aspects, the reception device may receive a third operation, and the imaging instruction device may instruct the imaging device to change an imaging direction, according to the third operation.

(11) In the imaging assistance apparatus according to a seventh aspect, in the first aspect, the laser head may be housed in a housing independent of the projection instruction device and the imaging instruction device.

(12) In the imaging assistance apparatus according to an eighth aspect, in any one of the second to fourth aspects, the laser head may be housed in a housing independent of the projection instruction device, the focusing instruction device, and the imaging instruction device.

(13) In the imaging assistance apparatus according to a ninth aspect, in any one of the second to

fourth aspects, the laser head and the reception device may be housed in a single housing.
(14) In the imaging assistance apparatus according to a tenth aspect, in the ninth aspect, the projection instruction device and the imaging instruction device may be housed in the single housing.
(15) In the imaging assistance apparatus according to an eleventh aspect, in the ninth aspect, the projection instruction device, the focusing instruction device, and the imaging instruction device may be housed in the single housing.
(16) In the imaging assistance apparatus according to a twelfth aspect, in any one of the first to eleventh aspects, the laser light source and the laser head are housed in a single housing.
(17) In the imaging assistance apparatus according to a thirteenth aspect, in any one of the first to twelfth aspects, the laser head may project the patterned light with one line or a cross-line in which a plurality of lines are crossed, as the pattern.
(18) In the imaging assistance apparatus according to a fourteenth aspect, in any one of the first to thirteenth aspects, the optical element may be any one of a laser line generator, a rod lens, a cylindrical lens, or a diffraction grating.
(19) In the imaging assistance apparatus according to a fifteenth aspect, in any one of the first to fourteenth aspects, an output of the laser light may be 0.39 mW to 1.0 mW.
(20) According to a sixteenth aspect, there is provided an imaging system comprising: the imaging assistance apparatus according to any one of the first to fifteenth aspects; and the imaging device, in which the imaging device performs the focusing control and the imaging in response to an instruction from the imaging assistance apparatus.
(21) In the imaging system according to a seventeenth aspect, in the sixteenth aspect, the imaging device may recognize the pattern in response to an instruction to perform the focusing control, and focus on a part on which the recognized pattern is projected.
(22) In the imaging system according to an eighteenth aspect, in the sixteenth or seventeenth aspect, a memory in which a repetition instruction for causing the imaging assistance apparatus to repeat a preparation operation performed before the first operation and the first operation at a predetermined cycle is recorded may be further provided, the imaging assistance apparatus may repeat an instruction to start the projection of the patterned light based on the preparation operation and to end the projection of the patterned light based on the first operation, an instruction to perform the focusing control, and an instruction to perform the imaging, in accordance with the repetition instruction, and the imaging device may repeat the focusing control and the imaging at the cycle in response to the instruction from the imaging assistance apparatus to image the subject.
(23) In the imaging system according to a nineteenth aspect, in any one of the sixteenth to eighteenth aspects, the imaging device may detect a plane region of the subject on the basis of the projected pattern and perform the focusing control on the plane region.
(24) In the imaging system according to a twentieth aspect, in any one of the sixteenth to nineteenth aspects, the imaging device may perform the focusing control on the basis of a signal output from a focusing control pixel disposed in an imaging element, and the imaging assistance apparatus may project the patterned light having a pattern corresponding to a disposition direction of the focusing control pixel.
(25) In the imaging system according to a twenty-first aspect, in any one of the sixteenth to twentieth aspects, the imaging device may perform the focusing control on the basis of a signal output from a focusing control pixel disposed in an imaging element, and the imaging assistance apparatus may project laser light having a wavelength range that at least partially overlaps with a transmission wavelength range of an optical filter disposed in the focusing control pixel, as the patterned light.
(26) According to a twenty-second aspect of the present invention, there is provided an imaging assistance method performed by a laser light projection device, which has a laser light source that generates laser light and a laser head provided with an optical element that projects the laser light

as patterned light having a determined pattern, and a control device, which has a processor, the method comprising: a reception step of receiving a first operation; a projection instruction step of instructing the laser light projection device during laser light projection to end the projection of the patterned light, on the basis of the first operation; and an imaging instruction step of instructing an imaging device to image a subject on which focusing control is performed, on the basis of the first operation. The twenty-second aspect may further include the same configurations as those of the second to twenty-first aspects.
(27) According to a twenty-third aspect of the present invention, there is provided an imaging assistance program causing a laser light projection device, which has a laser light source that generates laser light and a laser head provided with an optical element that projects the laser light as patterned light having a determined pattern, and a control device, which has a processor, to execute an imaging assistance method, the imaging assistance program causing the processor to execute: a reception function of receiving a first operation; a projection instruction output function of outputting an instruction to end the projection of the patterned light to the laser light projection device during laser light projection, on the basis of an input related to the received first operation; and an imaging instruction output function of outputting an instruction to image a subject on which focusing control is performed to an imaging device, on the basis of the input related to the first operation.
(28) As described above, with the imaging assistance apparatus, the imaging system, the imaging assistance method, and the imaging assistance program according to the aspects of the present invention, it is possible to satisfactorily perform focusing control by using patterned light.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram showing a schematic configuration of an imaging system according to a first embodiment.
(2) FIG. **2** is a diagram showing a functional configuration of a control device.
(3) FIGS. **3**A and **3**B are diagrams showing a configuration of a laser module.
(4) FIGS. **4**A, **4**B, and **4**C are views showing an example of an optical element.
(5) FIG. **5** is a view showing another example of the optical element.
(6) FIGS. **6**A and **6**B are views showing still another example of the optical element.
(7) FIGS. **7**A and **7**B are views showing a situation in which the laser module projects patterned light.
(8) FIG. **8** is a diagram showing a cross-line-shaped patterned light.
(9) FIG. **9** is a diagram showing a configuration of an imaging device.
(10) FIG. **10** is a diagram showing a functional configuration of an image processing device.
(11) FIG. **11** is a view showing a situation of imaging performed by the imaging system according to the first embodiment.
(12) FIGS. **12**A and **12**B are views showing a situation of plane recognition using patterned light and focusing control based on the recognition result.
(13) FIGS. **13**A and **13**B are diagrams showing an example of a sequence of the focusing control and imaging control.
(14) FIGS. **14**A and **14**B are diagrams showing another example of the sequence of the focusing control and the imaging control.
(15) FIG. **15** is a diagram showing a schematic configuration of an imaging system according to a second embodiment.
(16) FIG. **16** is a diagram showing a functional configuration of a control device according to the second embodiment.

(17) FIG. **17** is a view showing a situation of imaging performed by the imaging system according to the second embodiment.
(18) FIG. **18** is another view showing the situation of the imaging performed by the imaging system according to the second embodiment.
(19) FIG. **19** is a diagram showing a schematic configuration of an imaging system according to a third embodiment.
(20) FIG. **20** is a diagram showing a functional configuration of a control device according to the third embodiment.
(21) FIGS. **21**A and **21**B are views showing a situation of imaging performed by the imaging system according to the third embodiment.
DESCRIPTION OF THE PREFERRED EMBODIMENTS
(22) [Imaging of Social Structures]
(23) Regular inspections or reports are required for social structures, such as bridges, tunnels, roads, and buildings (sometimes called “social infra-structures”; “infra” is an abbreviation for “infrastructure”). Concrete and asphalt are often used as materials for these structures. For this reason, in a case where these structures are subjects, it is often difficult to focus because these structures have few features. In addition, in order to recognize damage, such as fine cracking, from the captured image, an illuminance of at least 320 lx (lx represents the unit of illuminance “lux”) is required, but the imaging location is often in a low illuminance environment (for example, the environmental illuminance under the bridge is very dark, about several tens of lux), and it is often difficult to focus. Therefore, in a case where the social structure is imaged, it is preferable to project auxiliary light in order to perform accurate focusing control.
(24) Further, in the case of the social structure, imaging is often repeated while changing the direction of the imaging device or moving the position of the imaging device, for reasons such as “long and big” and “provided in a high place”. Therefore, it is preferable that the imaging device or an auxiliary light projection device has a structure which is easy to mount on a panhead or a moving body (vehicle, robot, drone, or the like) and which is easy to handle.
(25) However, the conventional techniques do not sufficiently consider such circumstances. The imaging assistance apparatus, the imaging system, the imaging assistance method, and the imaging assistance program of the embodiment of the present invention have been made in view of such circumstances, and embodiments to be exemplified are as follows. A description will be given with reference to the accompanying drawings as necessary.
First Embodiment
(26) [Configuration of Imaging System]
(27) FIG. **1** is a diagram showing a schematic configuration of an imaging system **10** (imaging system) according to a first embodiment. The imaging system **10** comprises a remote release **100** (imaging assistance apparatus, reception device, and processor), a control device **200** (projection instruction device, focusing instruction device, imaging instruction device, and processor), an automatic panhead **300** (projection instruction device and processor), an automatic panhead **310** (focusing instruction device, imaging instruction device, and processor), a laser light projection device **400** (imaging assistance apparatus, laser light source, optical element, laser head, and laser light projection device), an imaging device **500** (imaging device and processor), and a strobe device **505**. The remote release **100**, the control device **200**, the laser light projection device **400**, and the imaging device **500** are housed in independent housings, respectively. The connection between these devices may be wired or wireless.
(28) [Configuration of Remote Release]
(29) The remote release **100** (imaging assistance apparatus, reception device, and processor) comprises an operation member **105** (see FIG. **11**) and a processor (not shown), and receives an operation (preparation operation and first operation) of a user via the operation member **105** (reception function). Each function of the remote release **100** (operation reception function,

projection instruction output function, focusing instruction function, projection end instruction function, and imaging instruction function) can be realized by using various processors, recording media, and electrical circuits, as in the control device **200**, which will be described later. As the operation member **105**, for example, a button or a switch that can be moved (pushed) in two stages within a determined stroke range can be used. The operation member **105** may be composed of a plurality of members. The remote release **100** can receive an operation to push the operation member **105** to the middle of the stroke as the preparation operation, and can receive an operation to push the operation member **105** to the end of the stroke as the first operation. The preparation operation is an operation to instruct a laser head **416** (laser head; see FIGS. **3**A and **3**B) to start the projection of patterned light (auxiliary light) and to instruct the imaging device **500** (imaging device) to perform focusing control on the subject on which the patterned light is projected. The first operation is an operation to instruct (projection end instruction) the laser light projection device **400** during laser light projection to end the projection of the patterned light and to instruct (imaging instruction) the imaging device **500** to image the subject on which the focusing control is performed. The instruction from the remote release **100** to the laser head **416** and the imaging device **500** is given via the control device **200**.
(30) In the remote release **100**, a lever-shaped (control stick-shaped) member or a handle-type member may be used as the operation member. Further, as the operation target, a device that is operated by, for example, voice or gesture, or a figure, a symbol, an icon, or the like that is electronically moved by a slide operation or the like on a touch panel may be used instead of a mechanical member.
(31) It is preferable that the remote release **100** is a device which receives an operation (second operation) to rotationally move the automatic panhead **300** and/or an operation (third operation) to rotationally move the automatic panhead **310** through the operation member **105** or the like, and which instructs the automatic panheads **300** and **310** to move rotationally via the control device **200**.
(32) [Configuration of Control Device]
(33) The control device **200** (imaging assistance apparatus, projection instruction device, focusing instruction device, imaging instruction device, and processor) gives instructions to the automatic panhead **300**, the automatic panhead **310**, the laser light projection device **400**, and the imaging device **500** in response to the instruction (the preparation operation and an instruction corresponding to the first operation) from the remote release **100**. FIG. **2** is a diagram showing a functional configuration of the control device **200**. As shown in FIG. **2**, the control device **200** has a projection instruction unit **212** (projection instruction output function), a focusing instruction unit **214** (focusing instruction output function), an imaging instruction unit **216** (imaging instruction output function), a communication control unit **218** (communication control function), and a panhead control unit **220** (panhead control function). The projection instruction unit **212** (projection instruction device and projection instruction output function) instructs laser modules **410** and **420** to start or end the projection of the patterned light, on the basis of an input related to the preparation operation and/or the first operation received by the remote release **100**, the focusing instruction unit **214** (focusing instruction device) instructs the imaging device **500** to perform the focusing control on the basis of the preparation operation, and the imaging instruction unit **216** (imaging instruction device) instructs the imaging device **500** to image the subject on which the focusing control is performed, on the basis of the input related to the first operation. The communication control unit **218** controls communication between the control device **200** and the remote release **100**, the automatic panhead **300**, the automatic panhead **310**, the laser light projection device **400**, and the imaging device **500**. The panhead control unit **220** instructs the automatic panheads **300** and **310** to move rotationally.
(34) The above-described functions of the control device **200** can be realized by using various processors and recording media. Examples of the various processors include a central processing

unit (CPU) which is a general-purpose processor that executes software (program) to realize various functions, a graphics processing unit (GPU) which is a processor specialized in image processing, and a programmable logic device (PLD) which is a processor of which the circuit configuration is changeable after manufacture, such as a field programmable gate array (FPGA). Each function may be realized by one processor, or may be realized by a plurality of processors of the same type or different types (for example, a plurality of FPGAs, a combination of a CPU and an FPGA, or a combination of a CPU and a GPU). Alternatively, a plurality of functions may be realized by one processor. More specifically, the hardware structure of these various processors is electrical circuits (circuitry) in which circuit elements, such as semiconductor elements, are combined.
(35) In a case where the above-described processor or electrical circuits execute software (program), a computer (for example, various processors and electrical circuits constituting the control device **200** and/or a combination thereof) readable code of the software to be executed is stored in a non-temporary recording medium (memory) such as a ROM, and the computer refers to the software. At the time of execution, information stored in a storage device is used as necessary. Further, at the time of execution, for example, a random access memory (RAM; memory) is used as a temporary storage area.
(36) The control device **200** can be realized by, for example, installing the imaging assistance program according to the embodiment of the present invention in a device (computer), such as a personal computer, a smartphone, or a tablet terminal.
(37) [Configuration of Laser Light Projection Device]
(38) As shown in FIG. **1**, the laser light projection device **400** comprises the laser modules **410** and **420**. FIGS. **3**A and **3**B are diagrams showing configurations of the laser modules **410** and **420**. In the example shown in FIG. **3**A, in the laser module **410**, a laser light source **412** (laser light source) that generates laser light and the laser head **416** (laser head) are housed in a single housing, but the laser light source **412** and the laser head **416** may be housed in housings independent of each other. In this case, a working portion (a portion mounted on the automatic panhead **300**) can be further miniaturized. Further, the laser modules **410** and **420** may be provided with a collimator that generates a parallel beam from laser light.
(39) The laser head **416** comprises an optical element **418** (optical element) that projects laser light as patterned light having a determined pattern. In the example shown in FIG. **3**A, the optical element projects line-shaped patterned light (optical axis L**2**) having a spread beam angle θ, and in the example shown in FIG. **3**B, the laser module **410** comprises an optical element **419**, and the optical element **419** projects line-shaped patterned light (optical axis L**2**) having a beam width W. The pattern of the patterned light may be one line for one laser head or a cross-line in which a plurality of lines are crossed (see FIGS. **7**A, **7**B, and **8**).
(40) The output of the laser light can be 0.39 mW to 1.0 mW (class 2 of JIS C 6802). Further, it is preferable that the laser light has a wavelength range that at least partially overlaps with the transmission wavelength range of an optical filter disposed in the focusing control pixels disposed in an imaging element **522** of the imaging device **500**. For example, in a case where the focusing control pixel is a G pixel (a pixel in which a green color filter is disposed), it is preferable that the laser light is green laser light (a wavelength range that at least partially overlaps with a wavelength range of 495 nm to 570 nm).
(41) As for the laser module, for example, a product having a housing diameter of about **9** mm and a housing length of about 26 mm is on the market.
(42) In the examples shown in FIGS. **3**A and **3**B, the same configuration can also be adopted for the laser module **420**.
(43) [Aspect of Optical Element]
(44) As the above-described optical element **418**, for example, any one of a laser line generator, a rod lens, a cylindrical lens, or a diffraction grating can be used.

(45) FIGS. **4**A, **4**B, and **4**C are views showing an example (laser line generator) of the optical element. FIG. **4**A is a perspective view, and a laser line generator **430** has a shape in which both sides of the cylinder are cut off diagonally. FIGS. **4**B and **4**C are a side view and a bottom view, respectively. As shown in FIGS. **4**A, **4**B, and **4**C, the laser line generator **430** can generate a line-shaped beam (patterned light) having a spread in a uniaxial direction. A spread angle β of the beam is determined according to the vertical angle α and the like of the laser line generator **430**. The dimensions of the laser line generator vary, but for example, products having a height H**1** of about 9 to 10 mm, a height H**2** of about 4 to 5 mm, and a diameter D of about 5 mm are on the market. The laser line generator is sometimes called a "laser line generator lens" or a "Powell lens".
(46) FIG. **5** is a view showing another example (rod lens) of the optical element. In the example of FIG. **5**, a rod lens **435** is cylindrical and has an optical axis L**3**. The rod lens **435** projects light incident from a direction perpendicular to the optical axis L**3** as a line-shaped beam (patterned light) having a spread in a uniaxial direction (into a plane perpendicular to the optical axis L**3**).
(47) FIGS. **6**A and **6**B are views showing still another example (cylindrical lens) of the optical element. FIG. **6**A is an example in which one cylindrical lens **440** projects a line-shaped beam (patterned light and sheet light) having a spread in a uniaxial direction, and FIG. **6**B is an example in which a lens group **450** composed of the cylindrical lenses **440** and **442** projects a parallel line-shaped beam (patterned light and sheet light).
(48) In addition to those shown in FIGS. **4**A to **6**B, a diffraction grating may be used as the optical element to project patterned light (for example, patterned light composed of a plurality of spots). It should be noted that one optical element or a plurality of optical elements may be used for the projection of the patterned light.
(49) [Examples of Laser Module and Patterned Light]
(50) FIGS. **7**A and **7**B are views showing a situation in which the laser module projects patterned light. FIG. **7**A is a view showing a situation in which a laser module **410**A projects patterned light having one line-shaped pattern, and FIG. **7**B is a view showing a situation in which a laser module **410**B projects cross-line-shaped patterned light in which a plurality of lines are crossed. As shown in FIGS. **7**A and **7**B, a laser module that projects one line-shaped patterned light may be used, or a laser module that projects cross-line-shaped patterned light may be used. One or more laser modules may be used in order to obtain the desired pattern.
(51) FIG. **8** is a diagram showing cross-line-shaped patterned light. A pattern **700** shown in FIG. **8** is composed of cross-lines **702** and **704** in which two lines are crossed (orthogonal). Such a pattern **700** may be formed, for example, by using four laser modules **410**A shown in FIG. **7**A or by using two laser modules **410**B shown in FIG. **7**B.
(52) It is preferable that the pattern of the patterned light projected by using the optical elements illustrated in FIGS. **4**A to **6**B and the like is a pattern corresponding to a disposition direction of the focusing control pixels disposed in the imaging element **522** (see FIG. **9**) of the imaging device **500**. For example, in a case where the focusing control pixels are disposed in the horizontal direction of the imaging element (or an imaging device main body), a pattern extending in the vertical direction (direction orthogonal to the disposition direction) is preferable. Since the pattern **700** illustrated in FIG. **8** is composed of a plurality of orthogonal lines, the pattern **700** has a component orthogonal to the disposition direction of the focusing control pixels regardless of the direction in which the focusing control pixels are disposed. Therefore, the imaging device **500** can recognize the pattern **700** to perform highly accurate focusing control.
(53) The automatic panhead **300** is an electric automatic panhead that rotationally moves the laser light projection device **400** in a biaxial direction (pan and tilt) in response to instructions from the remote release **100** and the control device **200**.
(54) [Configuration of Imaging Device]
(55) FIG. **9** is a diagram showing a configuration of the imaging device **500** (imaging device). The imaging device **500** is composed of an interchangeable lens **510** (optical system) and an imaging

device main body **520** (imaging device main body), and a subject image (optical image) is formed on the imaging element **522** by an imaging lens including a zoom lens **512** and a focus lens **516**, which will be described later. The interchangeable lens **510** and the imaging device main body **520** can be attached to and detached from each other via a mount (not shown). The lens device may be fixed to a camera body. The strobe device **505** is connected to the imaging device **500** via a communication I/F **536** (I/F: interface), and the strobe device **505** emits electronic flash light in response to an instruction from an image processing device **528** (an imaging control unit **540** and an electronic flash light control unit **544**). A general digital camera can be used as the imaging device **500**.
(56) The automatic panhead **310** is an automatic panhead that rotationally moves the imaging device **500** in the biaxial direction (pan and tilt) in response to an instruction from the control device **200**. An automatic panhead capable of instructing the imaging device **500** to perform focusing and/or imaging may be used.
(57) [Configuration of Interchangeable Lens]
(58) The interchangeable lens **510** comprises the zoom lens **512**, a stop **514**, the focus lens **516**, and a lens drive unit **518**. The lens drive unit **518** drives the zoom lens **512** and the focus lens **516** forward and backward in response to a command from the image processing device **528** (lens drive control unit **542** in FIG. **10**; processor) to perform zoom adjustment and focus adjustment (focusing control). The focusing control can be performed by, for example, an image plane phase difference method, but a contrast method may also be used. Further, the lens drive unit **518** controls the stop **514** (stop mechanism) in response to a command from the image processing device **528** to adjust exposure. Meanwhile, information, such as positions of the zoom lens **512** and the focus lens **516** and an opening degree of the stop **514**, is input to the image processing device **528**. The interchangeable lens **510** has an optical axis L**1**.
(59) [Configuration of Imaging Device Main Body]
(60) The imaging device main body **520** comprises the imaging element **522** (imaging element), an AFE **524** (AFE: Analog Front End), an A/D converter **526** (A/D: Analog to Digital), and an image processing device **528** (processor). Further, the imaging device main body **520** comprises an operation unit **530**, a recording device **532**, a monitor **534** (display device), and the communication I/F **536**. The imaging device main body **520** may have a shutter (not shown) that is used to shut off light transmitted through the imaging element **522**. The shutter may be a mechanical shutter or an electronic shutter. In the case of the electronic shutter, the charge accumulation period of the imaging element **522** is controlled by the image processing device **528** so that the exposure time (shutter speed) can be adjusted.
(61) The imaging element **522** comprises a light receiving surface in which a large number of light receiving elements are arranged in a matrix form. Then, the image of subject light transmitted through the zoom lens **512**, the stop **514**, and the focus lens **516** is formed on the light receiving surface of the imaging element **522**, and is converted into an electrical signal by each light receiving element. A color filter of red (R), green (G), or blue (B) is provided on the light receiving surface of the imaging element **522**, and a color image of the subject can be acquired on the basis of a signal of each color. Further, the focusing control pixels (for example, pixels used for focusing control of an image plane phase difference method) may be disposed in the imaging element **522** in a determined direction (for example, a horizontal direction and/or a vertical direction of the imaging device main body **520**). The imaging device **500** (the imaging control unit **540** and the lens drive control unit **542**) can perform the focusing control on the basis of a signal output from such a focusing control pixel. The focusing control pixel can be a green (G) pixel, but the present invention is not limited thereto.
(62) As the imaging element **522**, various photoelectric conversion elements such as a complementary metal-oxide semiconductor (CMOS) and a charge-coupled device (CCD) can be used. The AFE **524** performs noise removal, amplification, and the like of an analog image signal

output from the imaging element **522**, and the A/D converter **526** converts the captured analog image signal into a digital image signal with a gradation width. In the CMOS type imaging element, an AFE, an A/D converter, and a digital signal processing unit may be incorporated in a chip.
(63) [Configuration of Image Processing Device]
(64) FIG. **10** is a diagram showing a functional configuration of the image processing device **528**. The image processing device **528** has the imaging control unit **540**, the lens drive control unit **542**, the electronic flash light control unit **544**, and a communication control unit **546**. These functions can be realized by using various processors (electrical circuits) and recording media in the same manner as described above for the control device **200**.
(65) [Configuration of Operation Unit]
(66) The operation unit **530** includes various switches, buttons, dials, and the like, and the user can perform operations necessary for imaging a subject by using these devices. A touch panel type device is used, whereby the monitor **534** may be used as the operation unit.
(67) [Configuration of Storage Unit]
(68) The recording device **532** is composed of a non-temporary recording medium, such as various semiconductor memories, and a control unit thereof, and can record captured images and the like.
(69) [Monitor]
(70) The monitor **534** (display device) is, for example, a device such as a liquid crystal display, and can display information necessary for acquiring an image, a captured image, or the like. As described above, a touch panel type device may be used as the monitor **534**.
(71) [Imaging of Subject]
(72) FIG. **11** is a view illustrating a situation of imaging with the imaging system **10** according to the first embodiment. In the example shown in FIG. **11**, the subject is a bridge **900**, which is an aspect of a concrete structure (viewed from below; the same applies to the following embodiments), and the bridge **900** has members, such as a floor slab **910**, a main girder **912**, a cross-beam **914**, a lateral brace **916**, and a sway brace **918**.
(73) [Preparation Operation and Projection Instruction and Focusing Instruction]
(74) In a case where the user performs an operation (preparation operation) to push the operation member **105** to the middle of the stroke, the remote release **100** receives the preparation operation (reception step) and instructs the laser light projection device **400** to start the projection of patterned light, according to the preparation operation (projection instruction step). In response to this instruction, the laser modules **410** and **420** start the projection of patterned light, and the pattern **700** is formed on the bridge **900** (floor slab **910**) which is the subject (projection step). The user can confirm that the pattern **700** is captured in the visual field of the imaging device **500**, through the monitor **534**. Further, the remote release **100** instructs the imaging device **500** to perform focusing control on the subject on which the patterned light is projected, according to the preparation operation (focusing instruction step). The instruction from the remote release **100** is once transmitted to the control device **200** and is finally given from the control device **200** to the laser light projection device **400** and/or the imaging device **500** (the same applies to the following processing).
(75) The imaging device **500** (the imaging control unit **540** and the lens drive control unit **542**) recognizes the pattern **700** in response to the instruction to perform the focusing control, drives the focus lens **516** via the lens drive unit **518**, and focuses on a part (a part of the floor slab **910** in this example) on which the pattern **700** is projected (focusing control step).
(76) At the time of imaging, the user can perform an operation (the second operation and the third operation) to rotationally move the automatic panheads **300** and **310**, with respect to the remote release **100**, as necessary. The remote release **100** receives this operation (reception step) and gives an instruction to rotationally move the automatic panheads **300** and **310** (via the projection instruction unit **212**, the imaging instruction unit **216**, and the panhead control unit **220** of the

control device **200**) (a projection direction change step and an imaging direction change step), and the automatic panheads **300** and **310** change the projection direction of the patterned light (a pattern formation position and a focusing position) and/or the imaging direction in response to this instruction.
(77) The start (end) of the projection of laser light may be performed by the laser light source **412** starting (ending) the generation of the laser light, or by providing a member that blocks laser light in the laser head **416** to stop (start) blocking the laser light by using this member.
(78) [Detection of Plane Region Based on Patterned Light]
(79) In the above-described focusing control, the imaging device **500** (the imaging control unit **540** and the lens drive control unit **542**) may detect the plane region on the basis of the projected pattern, and may perform the focusing control on the detected plane region, as illustrated in FIGS. **12**A and **12**B (views showing the situation of plane recognition using patterned light and the focusing control based on the recognition result) (focusing control step). FIG. **12**A shows a situation in which the pattern **700** is projected onto the floor slab **910** provided with the main girder **912**, and the pattern **700** is uneven at the portion of the main girder **912**. The imaging device **500** detects a plane region **910**A (a region between the main girders **912**), as shown in FIG. **12**B, from the shape of such a pattern **700** and focuses on the plane region **910**A. Such control is performed, so that it is possible to prevent the unnecessary structure (the main girder **912** in the example of FIGS. **12**A and **12**B) from being focused, and to acquire an image in which the plane (plane region **910**A) of the subject is accurately focused.
(80) [First Operation and Imaging Instruction]
(81) In a case where the user performs an operation (first operation) to push the operation member **105** to the end of the stroke following the preparation operation (that is, in a case where the preparation operation is received before the first operation), the remote release **100** receives the first operation (reception step) and instructs the laser modules **410** and **420** to end the projection of the patterned light, according to the first operation (projection instruction step). In response to this instruction, the laser modules **410** and **420** end the projection of the patterned light. Further, the remote release **100** instructs the imaging device **500** to image the subject on which the focusing control is performed, according to the first operation (imaging instruction step). The imaging device **500** (imaging control unit **540**) images the subject in response to this instruction (imaging step). In a case where the illuminance is low, the imaging device **500** causes the electronic flash light source of the strobe device **505** to emit light via the electronic flash light control unit **544** (light emission step).
(82) [Sequence of Focusing Control and Imaging Control]
(83) FIGS. **13**A and **13**B are diagrams showing an example of a sequence of the focusing control and the imaging control performed by the control device **200**. In FIG. **13**A, the upper part shows a signal (instruction) that is received from the remote release **100**, and the lower part shows a signal (instruction) that is output to the imaging device **500**. In the example of FIGS. **13**A and **13**B, the preparation operation is performed in the period from time T**1** to time T**2**, and in response to this, an instruction to start the projection of patterned light and to perform the focusing control is given by a signal S**1**. Further, the first operation is performed in the period from time T**2** to time T**3**, and in response to this, an instruction to end the projection of the patterned light and to perform imaging is given by the signal S**2**. The imaging device **500** maintains the focused state even in the period from time T**2** to time T**3**.
(84) Regarding FIG. **13**A, in the above-described sequence, the patterned light is projected only during the focusing control (from time T**1** to time T**2**), but the patterned light may be continuously output, and the focusing control may be continued during that period. Specifically, as shown in FIG. **13**B, the patterned light is not projected during the period from time T**2** to time T**3** (imaging period), and the patterned light is projected in the other period (until time T**2** and after time T**3**). In this mode, the patterned light is projected even in the period when the preparation operation is not

performed, and the focusing control (AF search; AF means "Auto Focus") is executed. As a result, it is not necessary to perform the focusing control from a state in which the degree of focusing is low, so that the focusing control is speeded up and the stability thereof is improved. Such a control sequence is effective in a case where the imaging device **500** is mounted on a moving body with a motion (drone, vehicle, or the like) or the like. Such a control mode may be called "pre-AF mode".
(85) [Control Based on Repetition Instruction]
(86) The projection of the patterned light, the focusing control, and the imaging described above can be repeated at a predetermined cycle. In a case where these are repeatedly performed, the repetition instruction to repeat the preparation operation performed before the first operation and the first operation at a predetermined cycle is recorded in the memory, and the remote release **100** and/or the control device **200** repeats the instruction to start the projection of the patterned light based on the preparation operation, the instruction to end the projection of the patterned light based on the first operation, the instruction to perform the focusing control, and the instruction to perform the imaging, in accordance with this repetition instruction. The imaging device **500** repeats the focusing control and the imaging at the above-described cycle in response to instructions from the remote release **100** and/or the control device **200** to image the subject. This imaging may be for moving images or continuous still images. For example, in the case of the sequence shown in FIG. **13**A, the cycle is ΔT (=T**3**−T**1**). The memory that records the repetition instruction can be provided in the remote release **100** and/or the control device **200**.
(87) [Control Based Only on First Operation]
(88) Although FIGS. **13**A, **13**B, and the like illustrate a case where the projection of the patterned light and the focusing control are performed according to the preparation operation performed before the first operation and the first operation is performed following the preparation operation, the start and end of the projection of the patterned light, the focusing control, and the imaging control can be performed even in a case where the first operation is performed without the preparation operation. Specifically, as shown in FIG. **14**A, in a case where the first operation is performed without the preparation operation (time T**1**), the remote release **100** and/or the control device **200** (the projection instruction device, the focusing instruction device, and the imaging instruction device) instructs the laser light projection device **400** to start the projection of the patterned light (time T**1**) and instructs the imaging device **500** to start the focusing control on the subject on which the patterned light is projected (time T**1**), on the basis of the first operation. Further, the remote release **100** and/or the control device **200** instructs the imaging device **500** to start the focusing control, and then instructs the laser light projection device **400** to end the projection of the patterned light (time T**2**) at a fixed interval (determined interval: T**2**−T**1**), and further instructs the imaging device **500** to image the subject on which the focusing control is performed (time T**2**). The remote release **100** and/or the control device **200** can set this "fixed interval" according to the characteristics of the imaging device **500** (focusing speed and the like), the characteristics of the subject, the imaging conditions, and the like.
(89) Further, the patterned light may be continuously projected as in the sequence shown in FIG. **13**B. Specifically, as shown in FIG. **14**B, the remote release **100** and/or the control device **200** instructs the laser light projection device **400** to project the patterned light until time T**2** and after time T**3**, and the projection of the patterned light can be stopped in the period from time T**2** to time T**3** (during the imaging period).
(90) [Effect of First Embodiment]
(91) As described above, with the imaging system **10** (the imaging assistance apparatus, the imaging system, the imaging assistance method, and the imaging assistance program) according to the first embodiment, the laser light projection device **400** and the imaging device **500** can be moved and rotationally moved independently of each other, and the control device **200** may not be necessarily moved or rotationally moved. Therefore, the working weight can be reduced, and the laser light projection device **400** and the imaging device **500** can be easily mounted on a moving

body (robot, vehicle, drone, or the like). Further, since the laser light projection device **400** and the imaging device **500** are independent of each other, the degree of freedom of the projection of patterned light is high, and the focusing control can be satisfactorily performed with the patterned light.

Second Embodiment

(92) A second embodiment of the present invention will be described. FIG. **15** is a diagram showing a schematic configuration of an imaging system **20** according to the second embodiment. In the above-described first embodiment, the remote release **100** and the control device **200** are housed in housings independent of each other, but in the second embodiment, the remote release **100**, a control device **210**, and the laser modules **410** and **420** are housed in a single housing as an imaging assistance apparatus **110** (imaging assistance apparatus). The configurations of the remote release **100** and the laser modules **410** and **420** are the same as those of the first embodiment, and the user can project patterned light onto the subject by operating the operation member **105**.

(93) FIG. **16** is a diagram showing a functional configuration of the control device **210**. The control device **210** comprises a memory **222** (memory) in which the above-described repetition instruction is recorded. The function of the control device **210** can also be realized by using various processors (electrical circuits) as in the control device **200** according to the first embodiment.

(94) FIG. **17** is a view showing a situation of imaging performed by the imaging system **20**. As shown in FIG. **17**, in a case where the user performs an operation (preparation operation) to push the operation member **105** of the imaging assistance apparatus **110** to, for example, the middle of the stroke, the remote release **100** (reception device) receives the preparation operation (reception step), and in response to this, the control device **210** instructs the laser modules **410** and **420** to start the projection of the patterned light (projection instruction step), and the pattern **700** is formed on the floor slab **910** (subject). The user can change the direction of the imaging assistance apparatus **110** to form the pattern **700** at the desired position.

(95) It is preferable that the imaging direction of the imaging device **500** can be changed as necessary. Therefore, it is preferable that the remote release **100** is a device that receives an operation to rotationally move the automatic panhead **310** (the above-described third operation), and it is preferable that the control device **210** (the imaging instruction unit **216** and the panhead control unit **220**) gives an instruction to rotationally move the automatic panhead **310**, according to the third operation.

(96) Further, the control device **210** instructs the imaging device **500** to perform the focusing control on the floor slab **910** on which the pattern **700** is projected (focusing instruction step), according to the preparation operation. With this, the focusing control is performed (focusing control step). In a case where the user performs an operation (first operation) to push the operation member **105** to, for example, the end of the stroke in a state in which the focusing control is performed, the control device **210** instructs the laser modules **410** and **420** to end the projection of the patterned light (projection instruction step), and the pattern **700** disappears as shown in FIG. **18**. The control device **210** also instructs the imaging device **500** to image the subject, according to the first operation (imaging instruction step). At this time, the imaging device **500** causes the electronic flash light source of the strobe device **505** to emit light as necessary (light emission step). The sequence of these controls can be performed in the same manner as in the examples of FIGS. **13**A and **13**B. Further, such a sequence is performed in accordance with the repetition instruction recorded in the memory **222**, so that the subject can be continuously imaged (moving images or continuous still images).

(97) In a case where the first operation is performed without the preparation operation in the same manner as described above for the first embodiment, the start and end of the projection of the patterned light, the focusing control, and the imaging can be performed according to the first operation.

(98) [Effect of Second Embodiment]

(99) As described above, with the imaging system **20** according to the second embodiment, the laser modules **410** and **420** and the imaging device **500** are independent of each other, so that the projection position (focus position) of the pattern can be freely determined. Further, with this, the degree of freedom of the projection of the patterned light is high, and the focusing control can be satisfactorily performed with the patterned light.

Third Embodiment

(100) A third embodiment of the present invention will be described. FIG. **19** is a diagram showing a schematic configuration of an imaging system **30** according to the third embodiment. In the imaging system **30**, the remote release **100** and the control device **210** are housed in a single housing as the imaging assistance apparatus **110** (imaging assistance apparatus, projection instruction device, focusing instruction device, and imaging instruction device), and the laser modules **410** and **420** (laser light projection device and imaging assistance apparatus) are housed in a strobe device **507**. The strobe device **507** comprises an electronic flash light source **508**. FIG. **20** is a diagram showing a functional configuration of the image processing device **528** provided in the imaging device **500**, and the third embodiment is different from the first and second embodiments in that a patterned light control unit **548** (imaging assistance apparatus and projection instruction device) is provided.

(101) FIGS. **21**A and **21**B are views showing a situation of imaging performed by the imaging system **30** according to the third embodiment. In the case of the imaging system **30**, the user can perform the start and end of the projection of the patterned light, the focusing control, and the imaging by operating the operation member **105** (the preparation operation and the first operation), as in the first and second embodiments. For example, as shown in FIG. **21**A, the focusing control in a state in which the pattern **700** is projected and the imaging in a state in which the pattern **700** disappears as shown in FIG. **21**B can be performed.

(102) It is preferable that the remote release **100** is a device that receives an operation to rotationally move the automatic panhead **310** (the above-described third operation), and it is preferable that the control device **210** (the projection instruction unit **212**, the imaging instruction unit **216**, and the panhead control unit **220**) gives an instruction to rotationally move the automatic panhead **310** in response to the instructions from the remote release **100** corresponding to the second operation and the third operation.

(103) Even in such an imaging system **30**, the focusing control can be satisfactorily performed with patterned light, as in the first and second embodiments.

(104) The embodiments of the present invention have been described above, but the present invention is not limited to the above-described aspects, and various modifications can be made without departing from the spirit of the present invention.

EXPLANATION OF REFERENCES

(105) **10**, **20**, **30**: imaging system
(106) **100**: remote release
(107) **105**: operation member
(108) **110**: imaging assistance apparatus
(109) **200**, **210**: control device
(110) **212**: projection instruction unit
(111) **214**: focusing instruction unit
(112) **216**: imaging instruction unit
(113) **218**: communication control unit
(114) **220**: panhead control unit
(115) **222**: memory
(116) **300**, **310**: automatic panhead
(117) **400**: laser light projection device
(118) **410**, **410**A, **410**B, **420**: laser module

(119) **412**: laser light source
(120) **416**: laser head
(121) **418**, **419**: optical element
(122) **430**: laser line generator
(123) **435**: rod lens
(124) **440**, **442**: cylindrical lens
(125) **450**: lens group
(126) **500**: imaging device
(127) **505**, **507**: strobe device
(128) **508**: electronic flash light source
(129) **510**: interchangeable lens
(130) **512**: zoom lens
(131) **514**: stop
(132) **516**: focus lens
(133) **518**: lens drive unit
(134) **520**: imaging device main body
(135) **522**: imaging element
(136) **524**: AFE
(137) **526**: A/D converter
(138) **528**: image processing device
(139) **530**: operation unit
(140) **532**: recording device
(141) **534**: monitor
(142) **536**: communication I/F
(143) **540**: imaging control unit
(144) **542**: lens drive control unit
(145) **544**: electronic flash light control unit
(146) **546**: communication control unit
(147) **548**: patterned light control unit
(148) **700**: pattern
(149) **702**, **704**: cross-line
(150) **900**: bridge
(151) **910**: floor slab
(152) **910**A: plane region
(153) **912**: main girder
(154) **914**: cross-beam
(155) **916**: lateral brace
(156) **918**: sway brace
(157) α: vertical angle
(158) β: spread angle
(159) D: diameter
(160) H**1**, H**2**: height
(161) L**1**, L**2**, L**3**: optical axis
(162) S**1**, S**2**: signal
(163) θ: beam angle
(164) T**1**, T**2**, T**3**: time
(165) W: beam width

Claims

1. An imaging system comprising: an imaging assistance apparatus comprising: a laser light projection device having a laser light source that generates laser light and a laser head provided with an optical element that projects the laser light as patterned light having a determined pattern; a reception device that receives a first operation; a projection instruction device that instructs the laser light projection device during laser light projection to end the projection of the patterned light, on the basis of the first operation; and an imaging instruction device that instructs an imaging device to image a subject on which focusing control is performed, on the basis of the first operation; and the imaging device, wherein the imaging device performs the focusing control and the imaging in response to an instruction from the imaging assistance apparatus, the imaging system further comprises a memory in which a repetition instruction for causing the imaging assistance apparatus to repeat a preparation operation performed before the first operation and the first operation at a predetermined cycle is recorded, the imaging assistance apparatus repeats an instruction to start the projection of the patterned light based on the preparation operation and to end the projection of the patterned light based on the first operation, an instruction to perform the focusing control, and an instruction to perform the imaging, in accordance with the repetition instruction, and the imaging device repeats the focusing control and the imaging at the cycle in response to the instruction from the imaging assistance apparatus to image the subject.
2. The imaging system according to claim 1, wherein the reception device receives a preparation operation before the first operation, the projection instruction device instructs the laser light projection device to start the projection of the patterned light, on the basis of the preparation operation, and the imaging assistance apparatus further comprises a focusing instruction device that instructs the imaging device to start the focusing control on the subject on which the patterned light is projected, on the basis of the preparation operation.
3. The imaging system according to claim 1, wherein the projection instruction device instructs the laser light projection device to start the projection of the patterned light, on the basis of the first operation, and the imaging assistance apparatus further comprises a focusing instruction device that instructs the imaging device to start the focusing control on the subject on which the patterned light is projected, on the basis of the first operation.
4. The imaging system according to claim 3, wherein the projection instruction device instructs the laser light projection device to end the projection of the patterned light at a fixed interval after the start of the projection of the patterned light, on the basis of the first operation, and the imaging instruction device instructs the imaging device to image the subject on which the focusing control is performed, after the focusing instruction device instructs the imaging device to start the focusing control, on the basis of the first operation.
5. The imaging system according to claim 1, wherein the reception device receives a second operation, and the projection instruction device instructs the laser light projection device to change a projection direction of the patterned light, according to the second operation.
6. The imaging system according to claim 1, wherein the reception device receives a third operation, and the imaging instruction device instructs the imaging device to change an imaging direction, according to the third operation.
7. The imaging system according to claim 1, wherein the laser head is housed in a housing independent of the projection instruction device and the imaging instruction device.
8. The imaging system according to claim 2, wherein the laser head is housed in a housing independent of the projection instruction device, the focusing instruction device, and the imaging instruction device.
9. The imaging system according to claim 2, wherein the laser head and the reception device are housed in a single housing.
10. The imaging system according to claim 9, wherein the projection instruction device and the imaging instruction device are housed in the single housing.

11. The imaging system according to claim 9, wherein the projection instruction device, the focusing instruction device, and the imaging instruction device are housed in the single housing.
12. The imaging system according to claim 1, wherein the laser light source and the laser head are housed in a single housing.
13. The imaging system according to claim 1, wherein the laser light source and the laser head are housed in housings independent of each other.
14. The imaging system according to claim 1, wherein the laser head projects the patterned light with one line or a cross-line in which a plurality of lines are crossed, as the pattern.
15. The imaging system according to claim 1, wherein the optical element is any one of a laser line generator, a rod lens, a cylindrical lens, or a diffraction grating.
16. The imaging system according to claim 1, wherein an output of the laser light is 0.39 mW to 1.0 mW.
17. The imaging system according to claim 1, wherein the imaging device recognizes the pattern in response to an instruction to perform the focusing control, and focuses on a part on which the recognized pattern is projected.
18. The imaging system according to claim 1, wherein the imaging device detects a plane region of the subject on the basis of the projected pattern and performs the focusing control on the plane region.
19. The imaging system according to claim 18, wherein the imaging device performs the focusing control on a region where the projected pattern is not uneven.
20. An imaging system comprising: an imaging assistance apparatus comprising: a laser light projection device having a laser light source that generates laser light and a laser head provided with an optical element that projects the laser light as patterned light having a determined pattern; a reception device that receives a first operation; a projection instruction device that instructs the laser light projection device during laser light projection to end the projection of the patterned light, on the basis of the first operation; and an imaging instruction device that instructs an imaging device to image a subject on which focusing control is performed, on the basis of the first operation; and the imaging device, wherein the imaging device performs the focusing control and the imaging in response to an instruction from the imaging assistance apparatus, the imaging device performs the focusing control on the basis of a signal output from a focusing control pixel disposed in an imaging element, and the imaging assistance apparatus projects the patterned light having a pattern corresponding to a disposition direction of the focusing control pixel.
21. An imaging system comprising: an imaging assistance apparatus comprising: a laser light projection device having a laser light source that generates laser light and a laser head provided with an optical element that projects the laser light as patterned light having a determined pattern; a reception device that receives a first operation; a projection instruction device that instructs the laser light projection device during laser light projection to end the projection of the patterned light, on the basis of the first operation; and an imaging instruction device that instructs an imaging device to image a subject on which focusing control is performed, on the basis of the first operation; and the imaging device, wherein the imaging device performs the focusing control and the imaging in response to an instruction from the imaging assistance apparatus, the imaging device performs the focusing control on the basis of a signal output from a focusing control pixel disposed in an imaging element, and the imaging assistance apparatus projects laser light having a wavelength range that at least partially overlaps with a transmission wavelength range of an optical filter disposed in the focusing control pixel, as the patterned light.
22. The imaging system according to claim 1, wherein the laser light projection device is capable of moving and rotationally moving, and the imaging device is capable of moving and rotationally moving independently of the laser light projection device.